第六章 气体动力循环

gas power cycles

本章主要内容

6-1 分析热力循环的一般方法

6-2 活塞式内燃机实际循环的简化

6-3 活塞式内燃机的理想循环

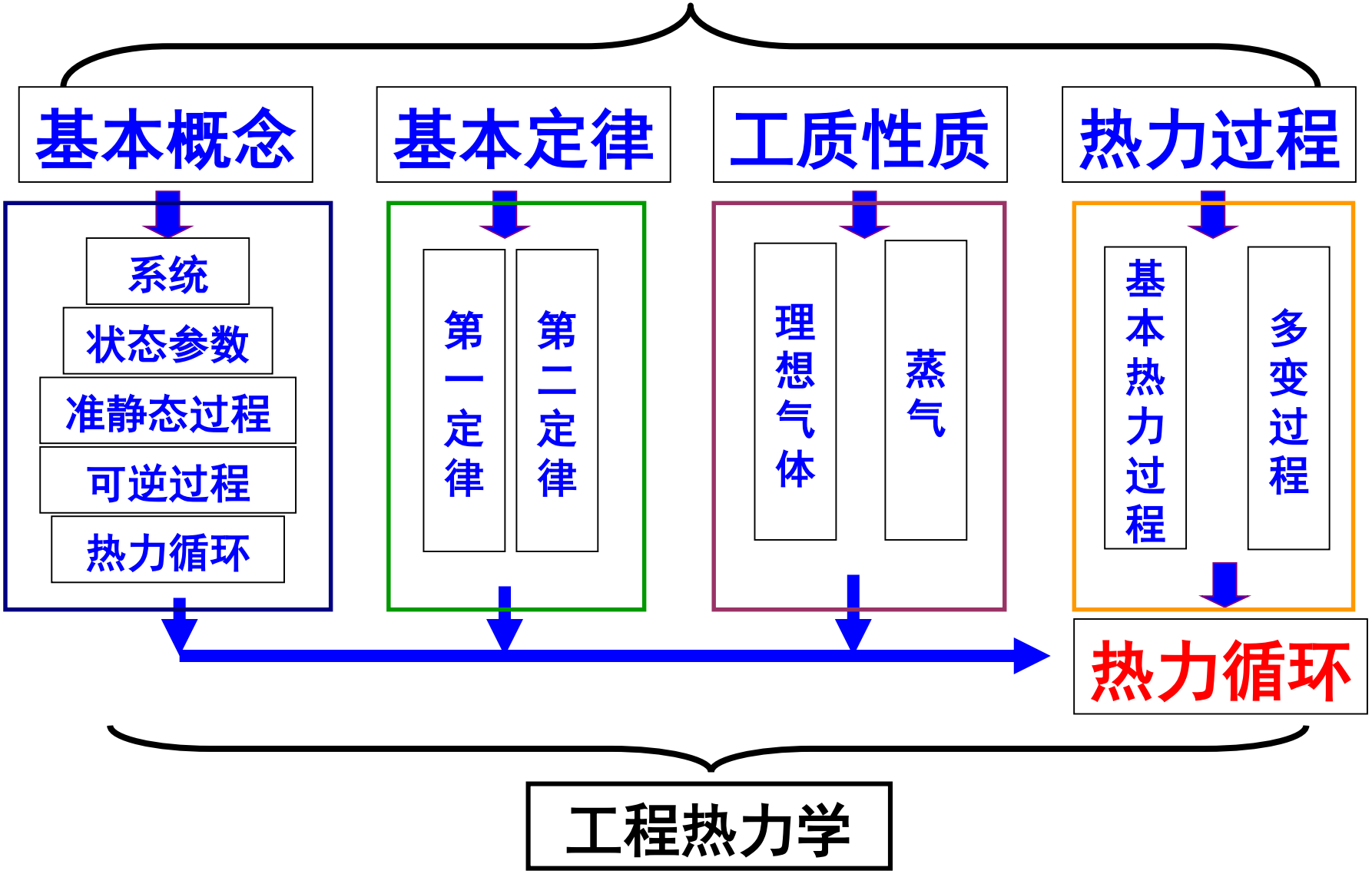
6-4 活塞式内燃机各种理想循环的热力学比较

6-5 燃气轮机装置循环

工程热力学

研究能量转换、特别是热能转化成机械能的规律和方法，以及提高转化效率的途径，以提高能源利用的经济性。

热能和机械能转换


基本概念
基本定律
工质性质
热力过程
系统
状态参数
准静态过程
可逆过程
热力循环
第一定律
第二定律
理想气体
蒸气
基本热力过程
多变过程
热力循环
工程热力学

热力循环

要实现连续作功，必须构成循环

定义：

工质从某一初态出发经历一系列热力状态变化后又回到原来初态的热力过程，即封闭的热力过程，叫热力循环。

热力循环

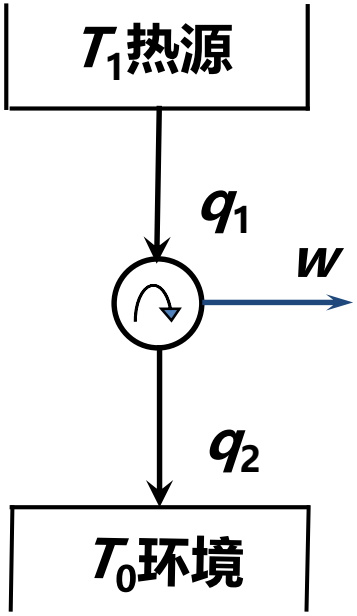
动力循环 ⇔ 正循环

制冷循环 ⇔ 逆循环

内燃机循环

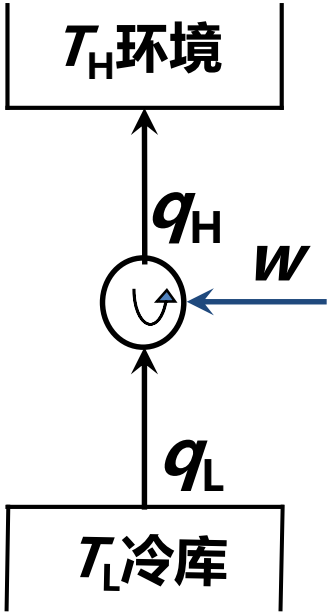
燃气轮机循环

蒸汽动力循环



制冷

制热



气体动力循环分类

按工质

- 气体动力循环：内燃机
  空气为主的燃气
  按理想气体处理
- 蒸汽动力循环：外燃机
  水蒸气等　实际气体

气体动力循环分类

按结构

- 活塞式 piston engine
  - 汽车，摩托，小型轮船
- 叶轮式 Gas turbine cycle
  - 航空，大型轮船，移动电站

气体动力循环分类

按燃料

- 汽油机 petrol (gasoline) engine

  小型汽车，摩托

- 柴油机diesel engine

  中、大型汽车，火车，轮船，移动电站

- 煤油机 kerosene oil engine

  航空

6–1　分析动力循环的一般方法

一、分析动力循环的目的和一般步骤

目的——在热力学基本定律的基础上分析循环能量转化的经济性，寻求提高经济性的方向及途径。

步骤：

1. 实际循环（复杂不可逆） $\xrightarrow{\text{抽象、简化}}$ **可逆理论循环**

$\xrightarrow{\text{分析可逆循环}}$ **影响经济性的主要因素和可能改进途径**

$\xrightarrow{\text{指导改善}}$ **实际循环**

2. 分析实际循环与理论循环的偏离程度，找出实际损失的部位、大小、原因及改进办法（主要由专业课承担）

二、分析动力循环的方法

1. 第一定律分析法 ——— 以第一定律为基础，以能量的数量守恒为立足点。以热效率为其指标。

2. 第二定律分析法 ——— 综合第一定律和第二定律，从能量的数量和质量分析。以做功能力损失和㶲效率为指标。

- 熵分析法 —— 熵产 —— 作功能力损失
- 㶲分析法 ——— 㶲损 ——— 㶲效率

6–2　活塞式内燃机实际循环的简化

一、活塞式内燃机 (internal combustion engine)简介

1．分类

按燃料：煤气机(gas engine)

汽油机(gasoline engine;　petrol engine)

柴油机(diesel engine)

按点火方式：点燃式(spark ignition engine)

压燃式(compression ignition engine)

按冲程：二冲程(two-stroke)

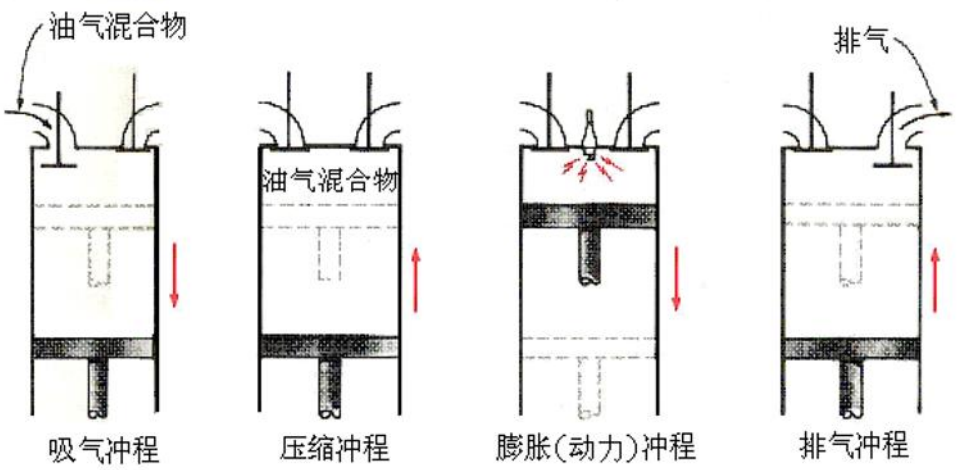
四冲程(four-stroke)

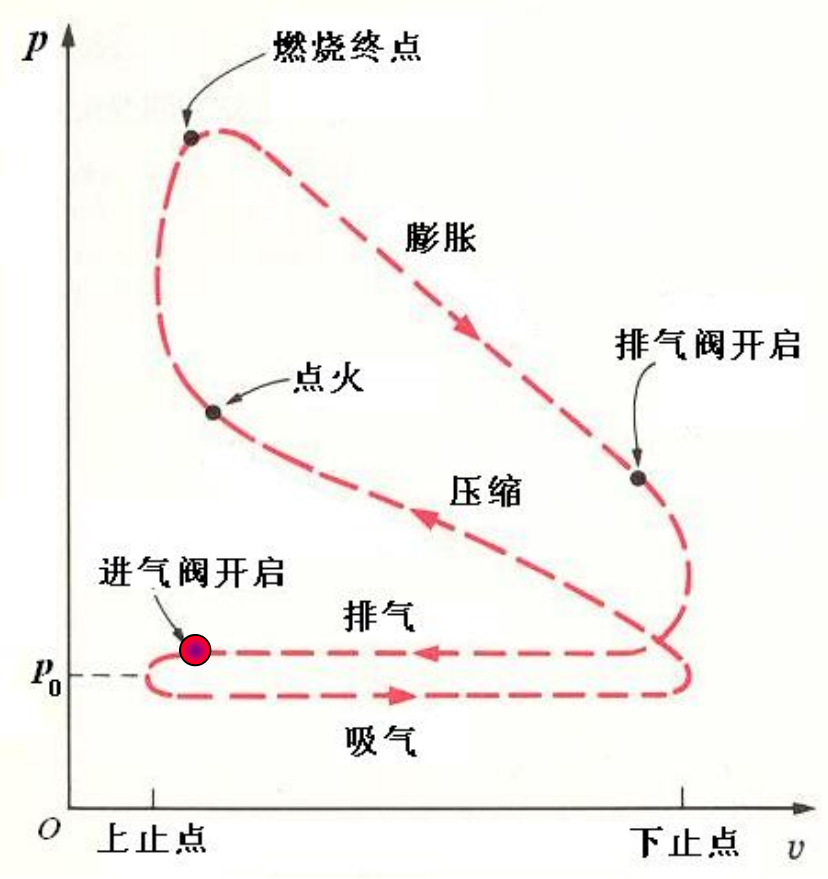
2．活塞式内燃机循环特点

开式循环(open cycle)；

燃烧、传热、排气、膨胀、压缩均为不可逆；

各环节中工质质量、成分稍有变化。





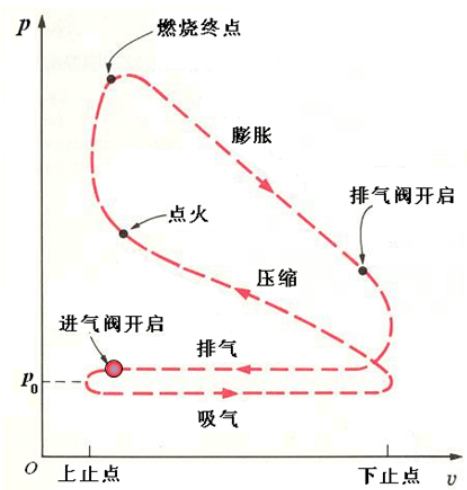
二、活塞式内燃机循环的简化

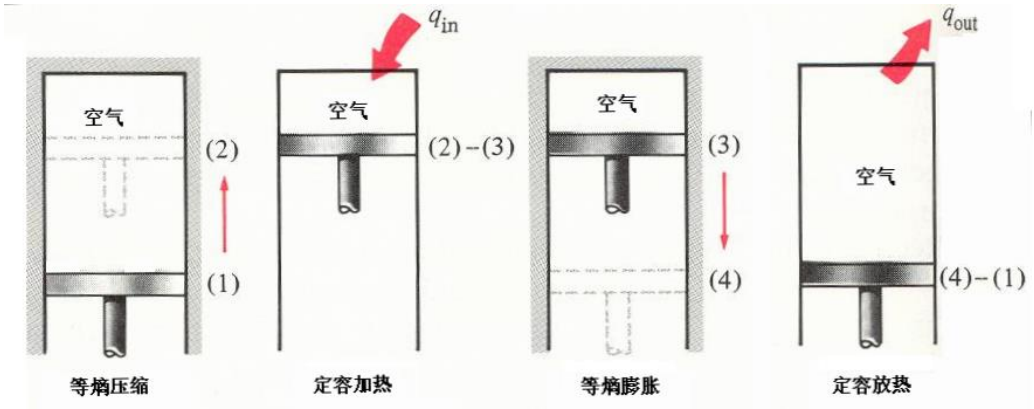
1. 空气标准假设(the air-standard hypothesis)

- 气体动力循环中工作流体 ⟶理想气体⟶ 空气⟶定比热
- 燃烧和排气过程 ⟶ 吸热和放热过程
- 燃料燃烧造成各部分气体成分及质量改变忽略不计

2. 活塞式内燃机循环简化

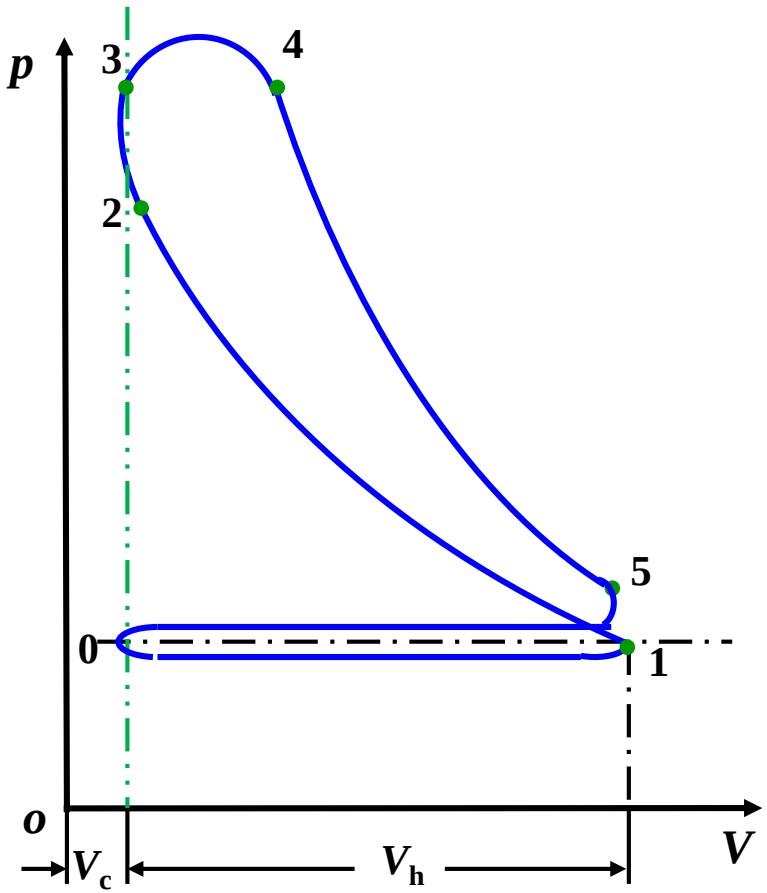
以汽油机为例

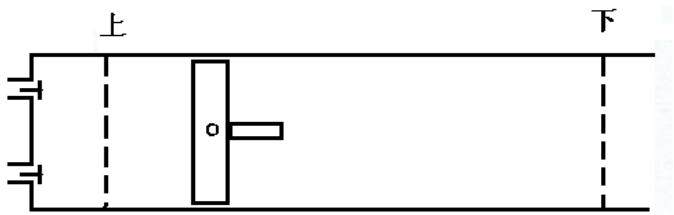




6–3 活塞式内燃机的理想循环

一、混合加热内燃机工作原理





0→1　吸气

1→2　压缩

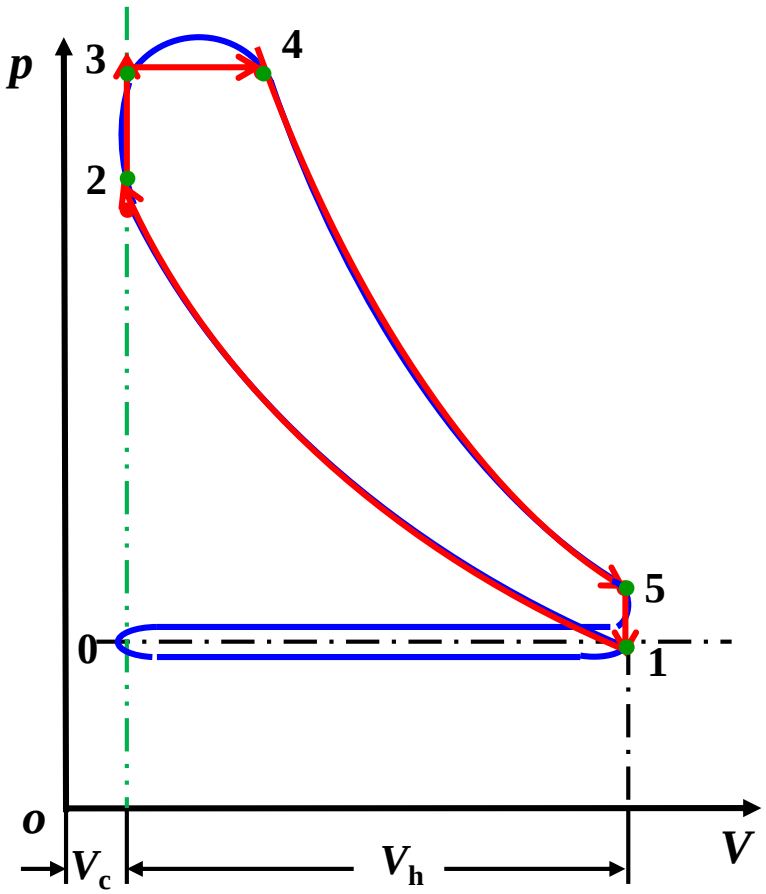
2→3　燃烧

3→4　燃烧

4→5　膨胀作功

5→0　排气

简化：引用空气标准假设



压缩→1-2等熵过程

燃烧→2-3等容吸热

燃烧→3-4定压吸热

膨胀→4-5等熵过程

排气→5-1等容放热

吸、排气线→重合、忽略

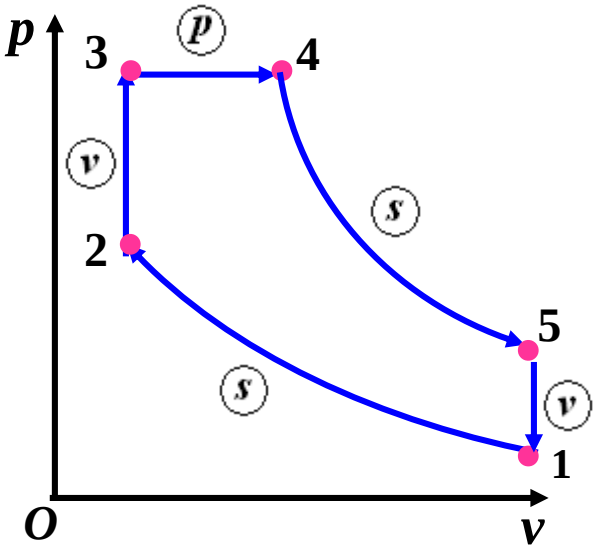
燃油质量→忽略

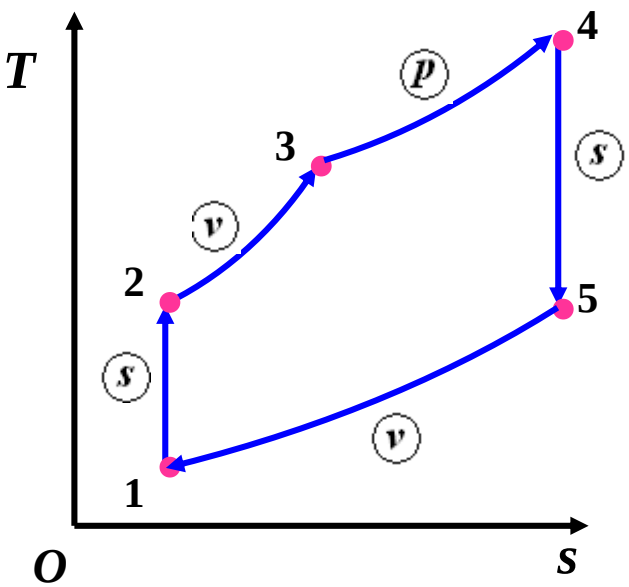
燃气成分改变→忽略

以空气为工质的定比热容内可逆闭式循环

二、混合加热理想循环（dual combustion cycle）

1. 混合加热理想循环的 *p-v*图及*T-s*图





1→2 等熵压缩；

2→3 等容吸热；

3→4 定压吸热；

4→5 等熵膨胀；

5→1 定容放热

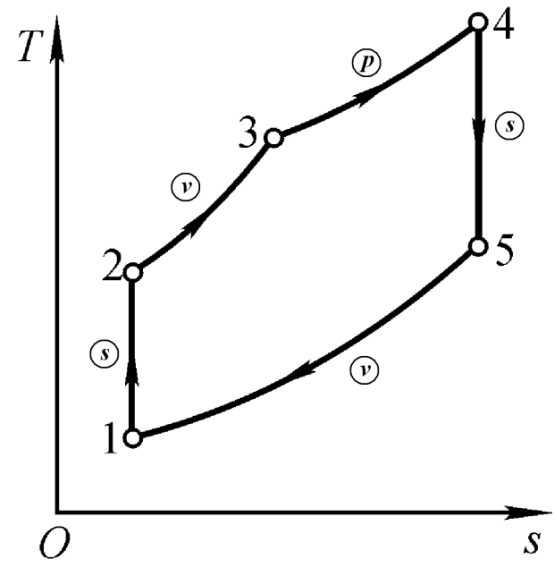
特性参数：

压缩比(compression ratio) $\varepsilon = \dfrac{v_1}{v_2}$

定容增压比(pressure ratio) $\lambda = \dfrac{p_3}{p_2}$

定压预胀比(cutoff ratio) $\rho = \dfrac{v_4}{v_3}$

2. 循环热效率



$$\eta_t = \frac{w_{net}}{q_1}$$

0

$$w_{net} = w_{1-2} + w_{2-3} + w_{3-4} + w_{4-5} + w_{5-1}$$

$$= w_{1-2} + w_{3-4} + w_{4-5}$$

$$= \frac{R_g}{\kappa-1} T_1 \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\kappa-1}{\kappa}} \right] + p_3 (v_4 - v_3) + \frac{R_g}{\kappa-1} T_4 \left[1 - \left(\frac{p_5}{p_4} \right)^{\frac{\kappa-1}{\kappa}} \right]$$

或 $w_{net} = q_{net} = q_1 - q_2$

$$q_1 = q_{2-3} + q_{3-4} = c_V (T_3 - T_2) + c_p (T_4 - T_3)$$

$$q_2 = q_{5-1} = c_V (T_5 - T_1)$$

$$\eta_t = 1 - \frac{q_2}{q_1} = 1 - \frac{T_5 - T_1}{(T_3 - T_2) + \kappa(T_4 - T_3)}$$

利用 ε、λ、ρ **表示** η_t

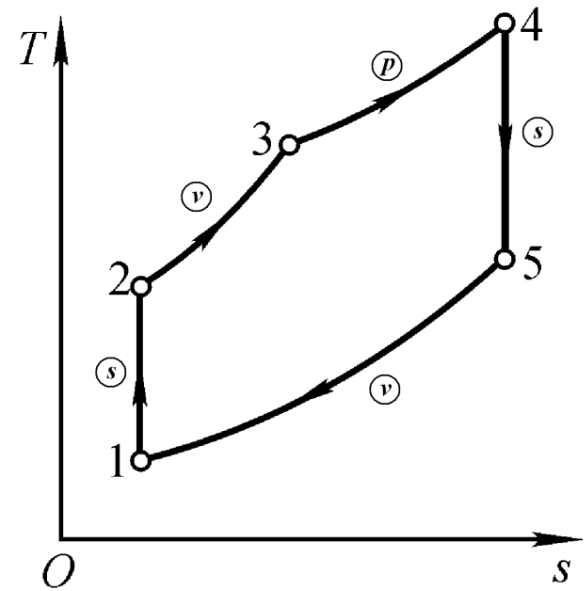
$1 \to 2$ $\quad T_2 = T_1 \left(\frac{v_1}{v_2}\right)^{\kappa-1} = T_1 \varepsilon^{\kappa-1}$

$\lambda = \frac{p_3}{p_2}$

$2 \to 3$ $\quad T_3 = T_2 \frac{p_3}{p_2} = T_1 \lambda \varepsilon^{\kappa-1}$

$3 \to 4$ $\quad T_4 = T_3 \frac{v_4}{v_3} = T_1 \rho \lambda \varepsilon^{\kappa-1}$

$5 \to 1$ $\quad T_5 = T_1 \frac{p_5}{p_1}$ $\quad \frac{p_5}{p_1}$?

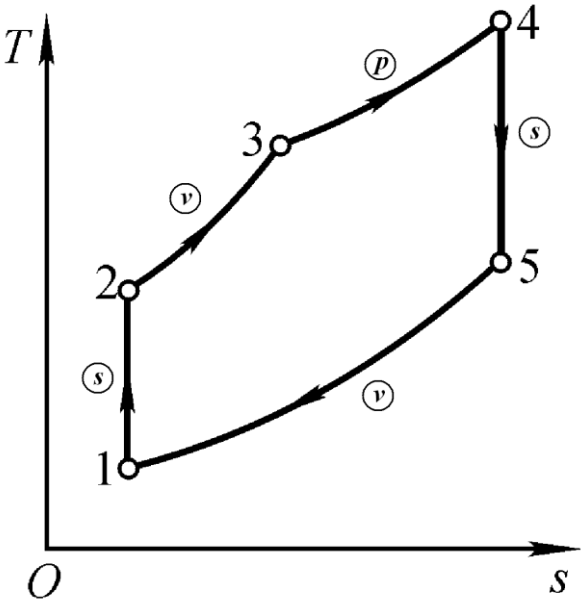


$$p_1 v_1^{\kappa} = p_2 v_2^{\kappa}; \quad p_5 v_5^{\kappa} = p_4 v_4^{\kappa}$$ **两式相除** $$\frac{p_5 v_5^{\kappa}}{p_1 v_1^{\kappa}} = \frac{p_4 v_4^{\kappa}}{p_2 v_2^{\kappa}}$$

$$v_1 = v_5$$

$$\frac{p_5}{p_1} = \frac{p_4}{p_2}\left(\frac{v_4}{v_2}\right)^{\kappa} = \frac{p_3}{p_2}\left(\frac{v_4}{v_3}\right)^{\kappa} = \lambda\rho^{\kappa}$$

$$p_4 = p_3 \quad v_2 = v_3$$



$$T_5 = T_1 \frac{p_5}{p_1} \qquad T_5 = T_1 \lambda \rho^{\kappa}$$

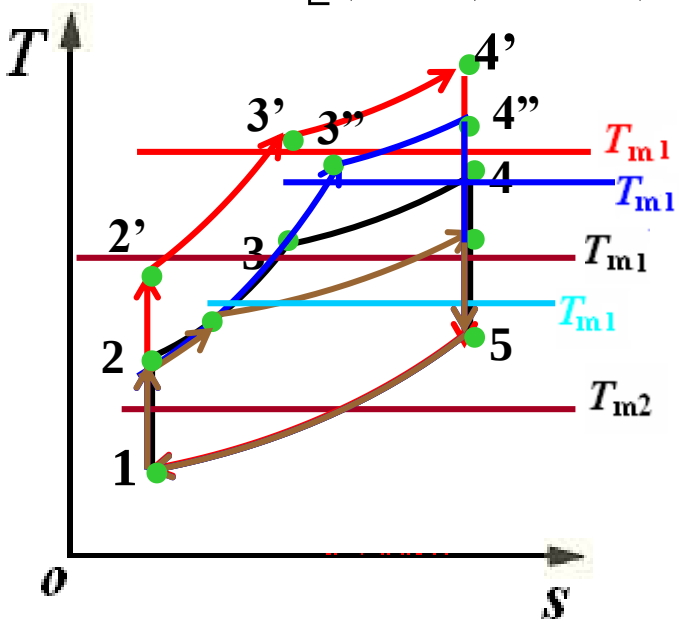
T_2、T_3、T_4和T_5代入

$$\eta_t = 1 - \frac{T_5 - T_1}{(T_3 - T_2) + \kappa(T_4 - T_3)}$$

$$\boxed{\eta_t = 1 - \frac{\lambda\rho^{\kappa} - 1}{\varepsilon^{\kappa-1}\left[(\lambda - 1) + \kappa\lambda(\rho - 1)\right]}}$$

讨论：

$$\eta_t = 1 - \frac{\lambda\rho^{\kappa} - 1}{\varepsilon^{\kappa-1}\left[(\lambda-1)+\kappa\lambda(\rho-1)\right]}$$



a) $\varepsilon \uparrow$ $\boxed{\eta_t \uparrow}$

$T_{m1}\uparrow$ T_{m2}不变

b) $\lambda \uparrow$ $\boxed{\eta_t \uparrow}$

$T_{m1}\uparrow$ T_{m2}不变

c) $\rho \uparrow$ $\boxed{\eta_t \downarrow}$

$T_{m1}\downarrow$ T_{m2}不变

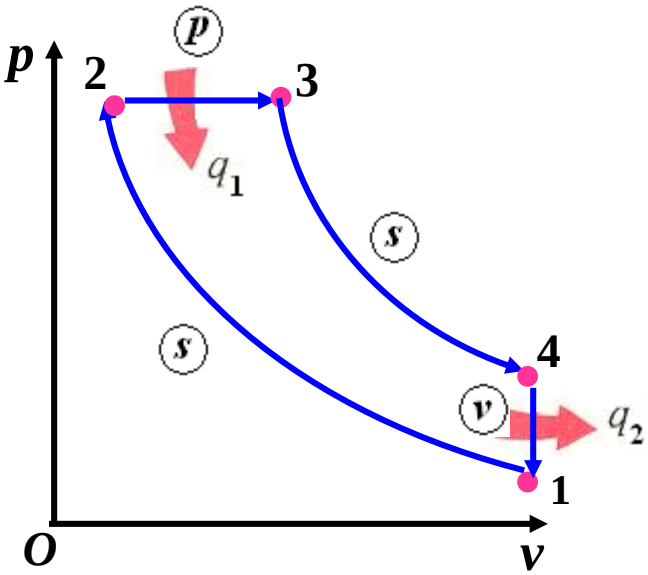
归纳：

（1）吸热前压缩气体，提高平均吸热温度是提高热效率的重要措施，是卡诺循环、第二定律对实际循环的指导。

（2）利用$T-s$图分析循环较方便。

（3）同时考虑 q_1和 q_2 或平均T_{1m} 和T_{2m}。

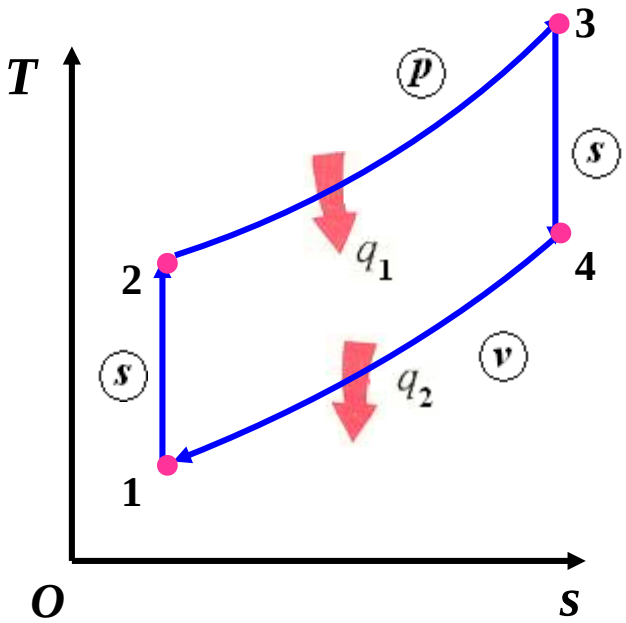
三、定压加热理想循环(Diesel cycle)



特性参数:

$$\varepsilon = \frac{v_1}{v_2}$$

$$\rho = \frac{v_3}{v_2}$$



$$\eta_t = 1 - \frac{q_2}{q_1}$$

$$q_2 = c_V\left(T_4 - T_1\right)$$

$$q_1 = c_p\left(T_3 - T_2\right)$$

$$\eta_t = 1 - \frac{T_4 - T_1}{\kappa\left(T_3 - T_2\right)}$$

$$\eta_t = 1 - \frac{\lambda\rho^{\kappa} - 1}{\varepsilon^{\kappa-1}\left[\left(\lambda - 1\right) + \kappa\lambda\left(\rho - 1\right)\right]}$$

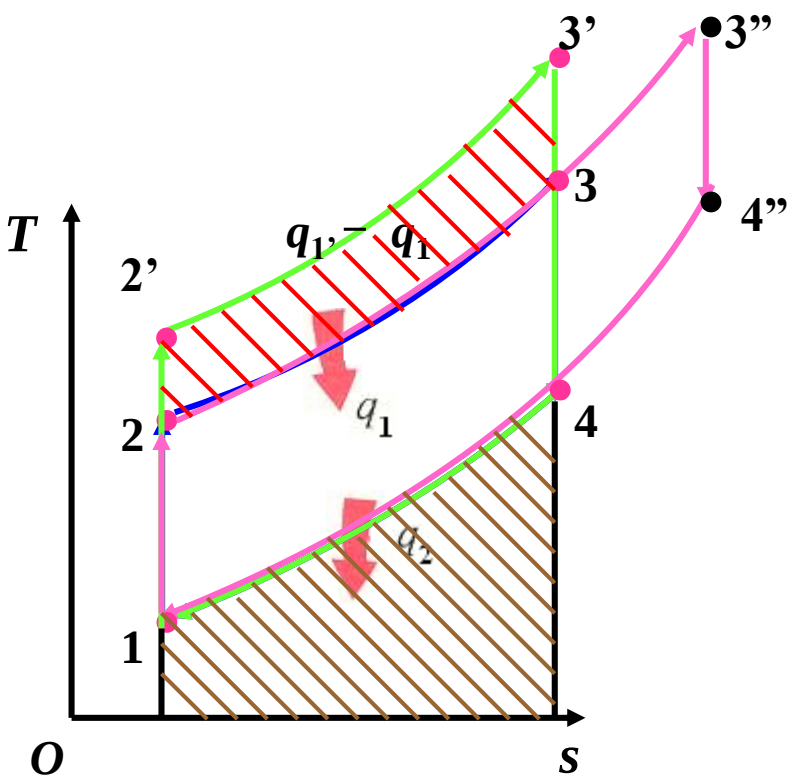
$$\lambda = 1 \Rightarrow \eta_t = 1 - \frac{\rho^{\kappa} - 1}{\kappa\varepsilon^{\kappa-1}\left(\rho - 1\right)}$$

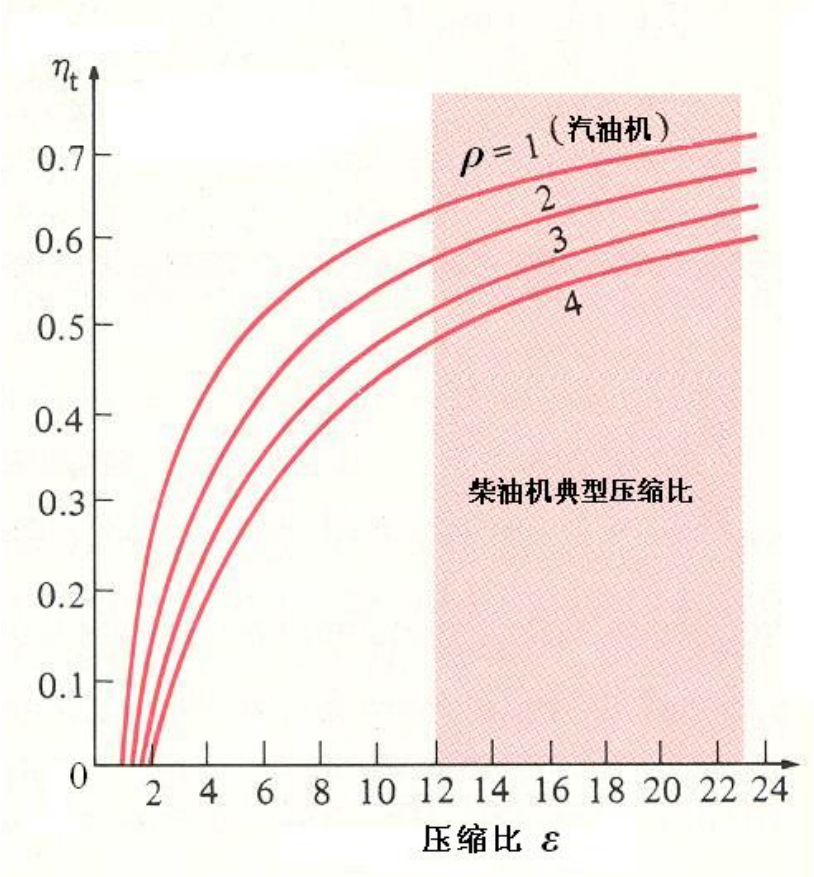
讨论：

$$\lambda = 1 \Rightarrow \eta_t = 1 - \frac{\rho^{\kappa} - 1}{\kappa \varepsilon^{\kappa-1}(\rho - 1)}$$

a) $\varepsilon \uparrow$ q_2不变，q_1增大 $\boxed{\eta_t \uparrow \quad w_{net} \uparrow}$

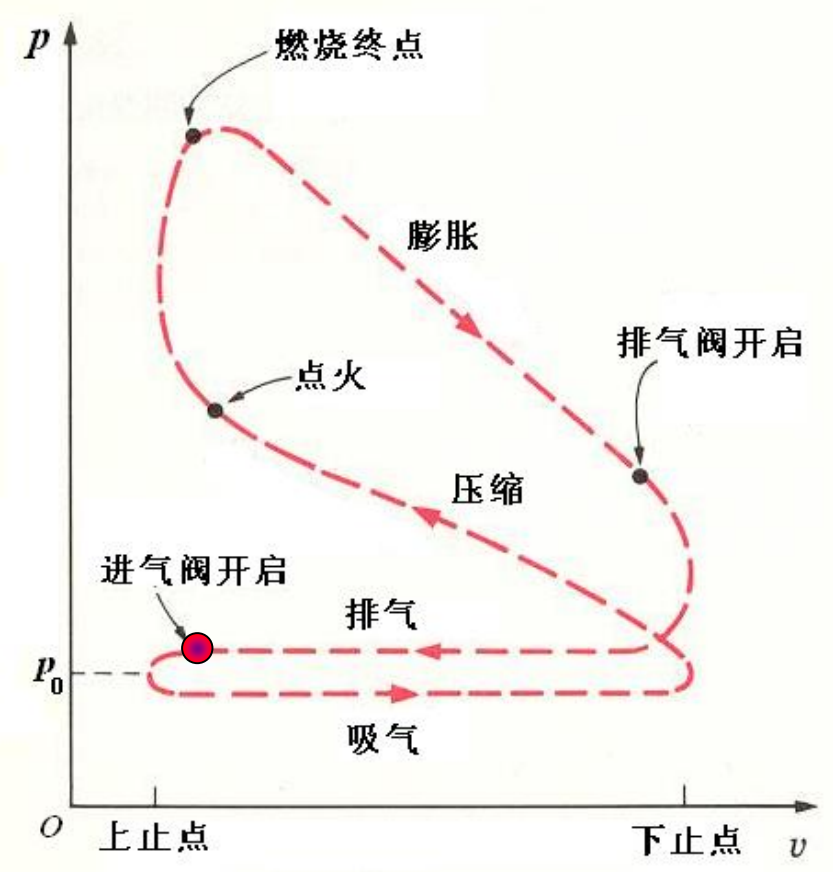
b) $\rho \uparrow$ $\boxed{\eta_t \downarrow \quad w_{net} \uparrow}$

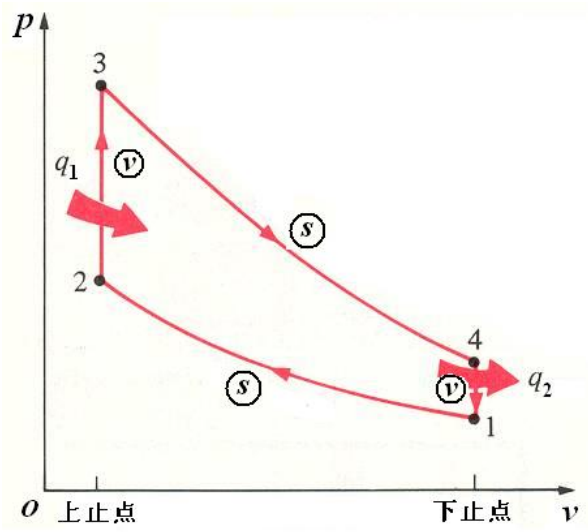


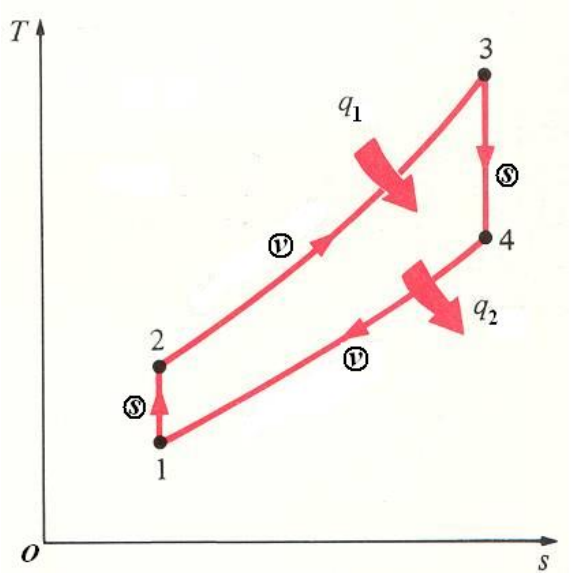


c) 重负荷（$\rho\uparrow$，$q_1\uparrow$）时内部热效率下降，除$\rho\uparrow$外还有因温度上升而使$\kappa\downarrow$，造成热效率下降。

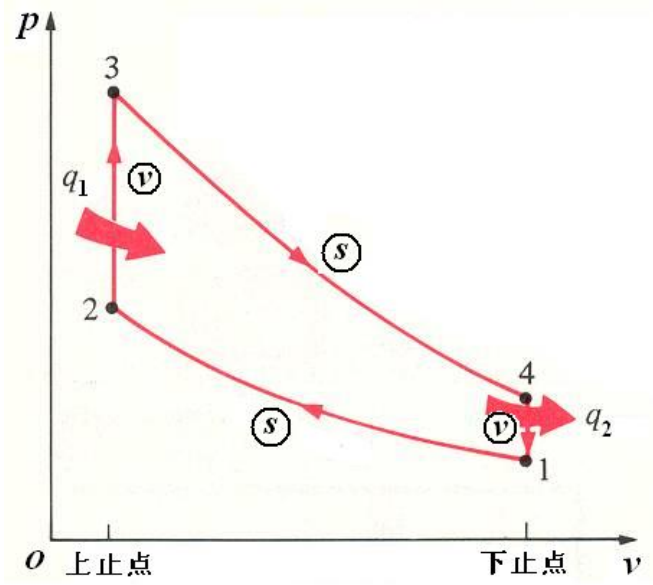
四、定容加热理想循环 (Otto cycle)

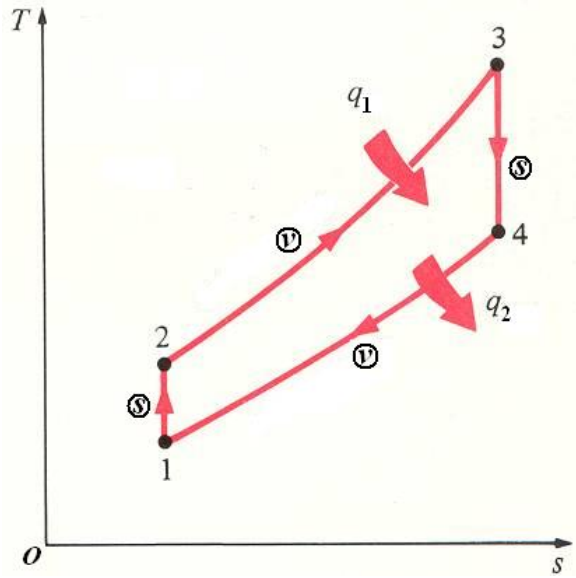






$$\varepsilon = \frac{v_1}{v_2} \qquad \lambda = \frac{p_3}{p_2}$$





$$\varepsilon = \frac{v_1}{v_2}$$

$$\lambda = \frac{p_3}{p_2}$$

$$q_1 = c_V\left(T_3 - T_2\right)$$

$$q_2 = c_V\left(T_4 - T_1\right)$$

$$\eta_t = 1 - \frac{q_2}{q_1} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

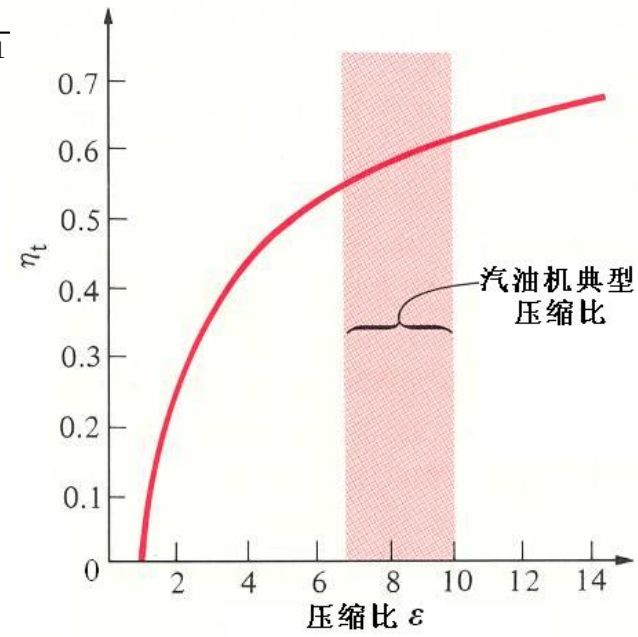
$$\rho = 1 \Rightarrow \eta_t = 1 - \frac{\lambda\rho^{\kappa} - 1}{\varepsilon^{\kappa-1}\left[\left(\lambda - 1\right) + \kappa\lambda\left(\rho - 1\right)\right]} = 1 - \frac{1}{\varepsilon^{\kappa-1}}$$

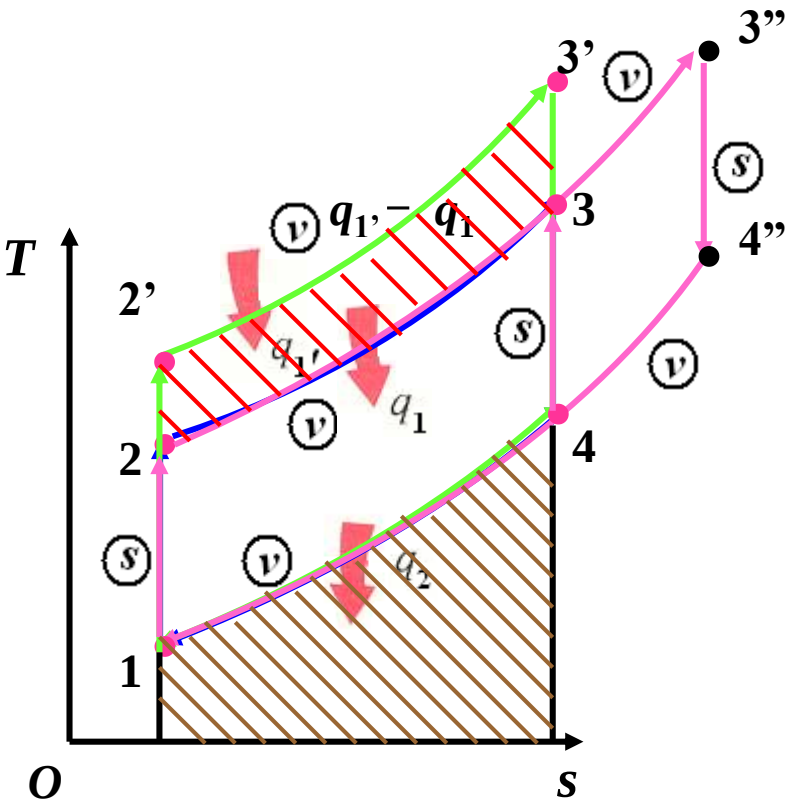
讨论： $\rho=1\Rightarrow\eta_t=1-\dfrac{\lambda\rho^{\kappa}-1}{\varepsilon^{\kappa-1}\left[(\lambda-1)+\kappa\lambda(\rho-1)\right]}=1-\dfrac{1}{\varepsilon^{\kappa-1}}$

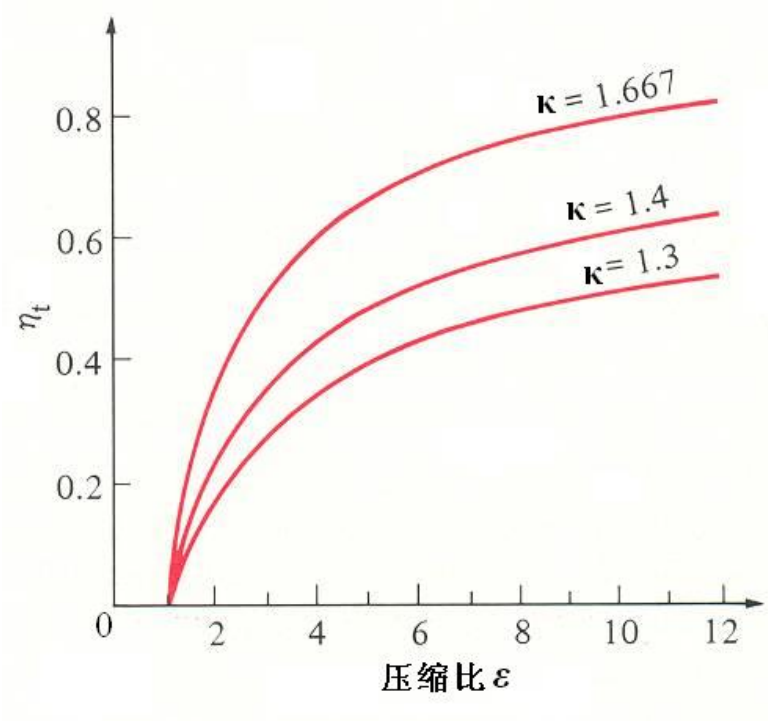
a) $\varepsilon\uparrow$ $\boxed{\eta_t\uparrow}$

b) $\lambda\uparrow$ η_t**不变** $w_{net}\uparrow$

c) **重负荷（$q_1\uparrow$）时内部热效率下降，因温度上升使 $\kappa\downarrow$，造成热效率下降**

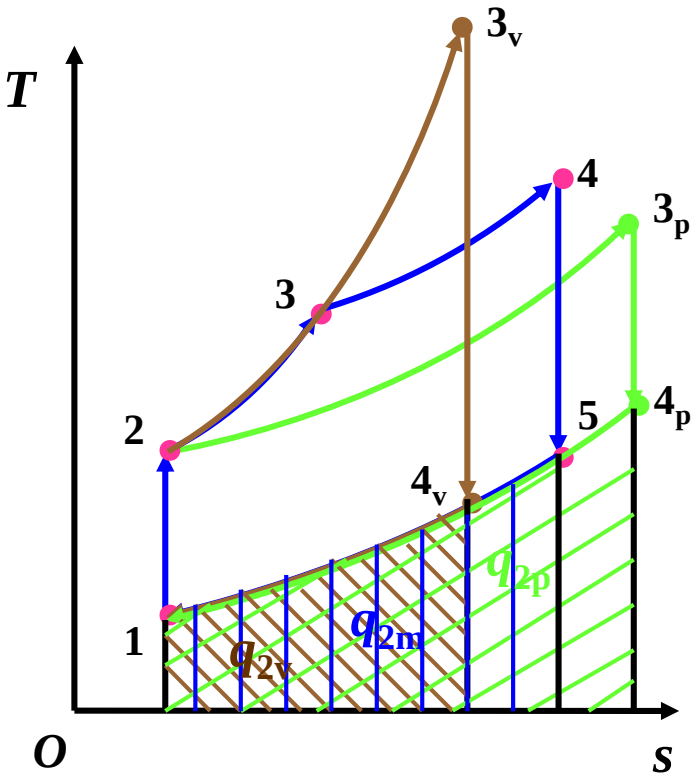




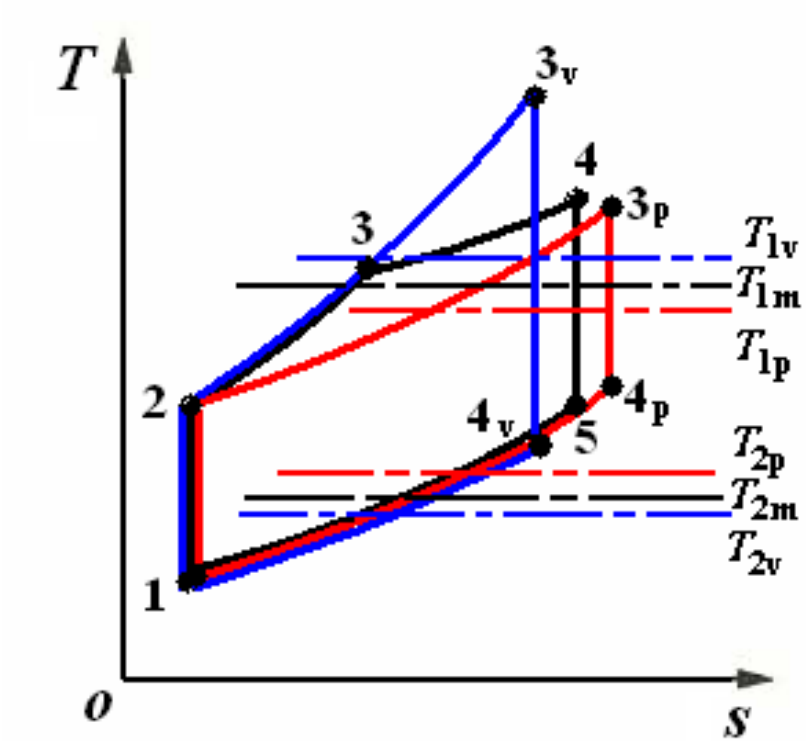


6–4 活塞式内燃机各种理想循环的热力学比较

一、压缩比相同，吸热量相同时的比较



或



$$q_{1v} = q_{1m} = q_{1p}$$

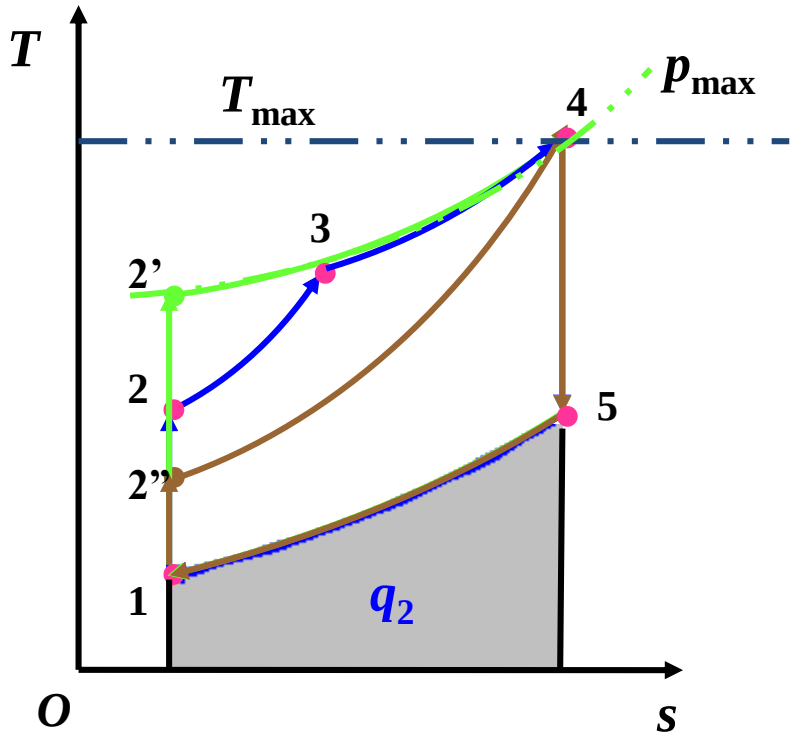
$$\overline{T}_{2p} > \overline{T}_{2m} > \overline{T}_{2v}$$

$$\eta_{tv} > \eta_{tm} > \eta_{tp}$$

$$q_{2v} < q_{2m} < q_{2p}$$

$$\overline{T}_{1p} < \overline{T}_{1m} < \overline{T}_{1v}$$

二、循环 p_{max}、T_{max} 相同时的比较



$$q_{2p} = q_{2m} = q_{2v}$$

$$q_{1p} > q_{1m} > q_{1v}$$

$$\eta_{t,p} > \eta_{t,m} > \eta_{t,v}$$

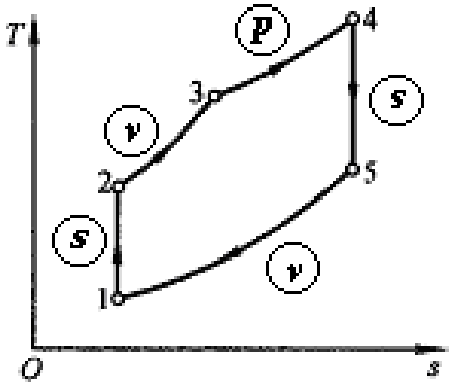
或

$$\overline{T}_{2p} = \overline{T}_{2m} = \overline{T}_{2v}$$

$$\overline{T}_{1p} > \overline{T}_{1m} > \overline{T}_{1v}$$

例题1：

已知柴油机混合加热理想循环p_1= 0.17 MPa、t_1= 60℃，压缩比ε =14.5，气缸中气体最大压力10.3 MPa，循环加热量q_1 = 900 kJ/kg。设工质为空气，比热容为定值并取 c_p = 1004 J/(kg·K)，c_V = 718 J/(kg·K)，κ = 1.4；环境温度t_0 = 20℃，压力p_0 = 0.1 MPa。试分析该循环并求循环热效率及㶲效率。

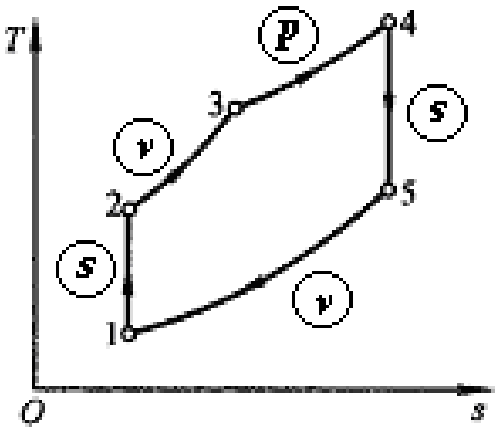


解　由已知条件 p_1= 0.17 kPa、T_1= 333.15 K

点1：
$$v_1=\frac{R_gT_1}{p_1}=\frac{287\ \text{J/(kg·K)}\times 333.15\ \text{K}}{0.17\times 10^6\ \text{Pa}}=0.562\ \text{m}^3/\text{kg}$$

点2：
$$v_2=\frac{v_1}{\varepsilon}=\frac{0.562\ \text{m}^3/\text{kg}}{14.5}=0.038\ 7\ \text{m}^3/\text{kg}$$

$$p_2 = \varepsilon p_1 = 14.5 \times 0.17\text{MPa} = 2.465\text{MPa}$$



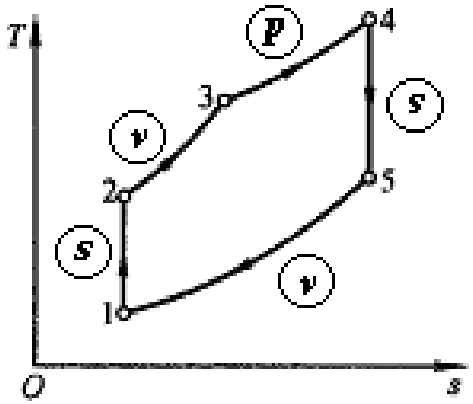
$$v_2 = v_1 \left(\frac{p_1}{p_2} \right)^{\kappa} = v_1 \left(\frac{1}{\varepsilon} \right)^{\kappa}$$

$$= 0.562\ \text{m}^3/\text{kg} \times \left(\frac{1}{14.5} \right)^{1.4} = 0.0133\ \text{m}^3/\text{kg}$$

$$T_2 = \frac{p_2 v_2}{R_g} = 114\text{K}$$?

$$\varepsilon = \frac{v_1}{v_2} \neq \frac{p_2}{p_1}$$

1-2是定熵过程，有



$$p_2 = p_1\left(\frac{v_1}{v_2}\right)^{\kappa} = p_1\varepsilon^{\kappa}$$

$$= 0.17\ \text{MPa} \times 14.5^{1.4} = 7.18\ \text{MPa}$$

$$T_2 = \frac{p_2 v_2}{R_g} = \frac{7.18\times10^6\ \text{Pa}\times 0.038\ 7\ \text{m}^3/\text{kg}}{287\ \text{J/(kg}\cdot\text{K)}} = 968\ \text{K}$$

点3： $p_3 = 10.3\ \text{MPa} \quad v_3 = v_2 = 0.038\ 7\ \text{m}^3/\text{kg}$

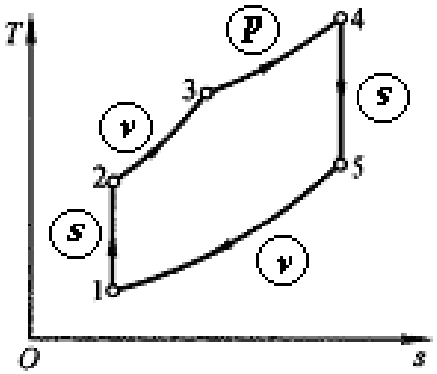
$$T_3 = \frac{p_3 v_3}{R_g} = \frac{10.3\times10^6\ \text{Pa}\times 0.038\ 7\ \text{m}^3/\text{kg}}{287\ \text{J/(kg}\cdot\text{K)}} = 1\ 389\ \text{K}$$

$$\lambda = \frac{p_3}{p_2} = \frac{10.3\ \text{MPa}}{7.18\ \text{MPa}} = 1.43$$

$$q_{1V} = c_V (T_3 - T_2)$$

$$= 0.718\ \text{kJ/(kg}\cdot\text{K)} \times (1\,389 - 968)\ \text{K}$$

$$= 302\ \text{kJ/kg}$$



$$q_{1p} = q_1 - q_{1V} = 900\ \text{kJ/kg} - 302\ \text{kJ/kg} = 598\ \text{kJ/kg}$$

点4: $p_4 = p_3 = 10.3\ \text{MPa}$，

$$q_{1p} = c_p (T_4 - T_3)$$

$$T_4 = T_3 + \frac{q_{1p}}{c_p} = 1\,389\ \text{K} + \frac{598\ \text{kJ/kg}}{1.004\ \text{kJ/(kg}\cdot\text{K)}} = 1\,985\ \text{K}$$

$$v_4 = \frac{R_g T_4}{p_4} = \frac{287\ \text{J/(kg}\cdot\text{K)} \times 1\,985\ \text{K}}{10.3 \times 10^6\ \text{Pa}} = 0.055\ \text{m}^3\text{/kg}$$

点5: $v_5 = v_1 = 0.562\ \text{m}^3\text{/kg}$

$$p_5 = p_4\left(\frac{v_4}{v_5}\right)^{\kappa} = 10.3\ \text{MPa}\times\left(\frac{0.055\ \text{m}^3/\text{kg}}{0.562\ \text{m}^3/\text{kg}}\right)^{1.4} = 0.398\ \text{MPa}$$

$$T_5 = \frac{p_5 v_5}{R_g} = \frac{0.398\times10^6\ \text{Pa}\times0.562\ \text{m}^3/\text{kg}}{287\ \text{J/(kg}\cdot\text{K)}} = 779\ \text{K}$$

$$q_2 = c_V(T_5 - T_1) = 0.718\ \text{kJ/(kg}\cdot\text{K)}\times(779-333)\ \text{K} = 320\ \text{kJ/kg}$$

$$w_{\text{net}} = q_1 - q_2 = 900\ \text{kJ/kg} - 320\ \text{kJ/kg} = 580\ \text{kJ/kg}$$

$$\eta_t = 1 - \frac{\lambda\rho^{\kappa} - 1}{\varepsilon^{\kappa-1}\left[(\lambda-1)+\kappa\lambda(\rho-1)\right]}$$

$$= 1 - \frac{1.43\times1.42^{1.4} - 1}{14.5^{1.4-1}\times\left[(1.43-1)+1.4\times1.43\times(1.42-1)\right]} = 0.639$$

或

$$\eta_t = \frac{w_{\text{net}}}{q_1} = \frac{580\text{kJ/kg}}{900\text{kJ/kg}} = 0.644$$

吸热过程中空气熵增

$$\Delta s_{2-4} = c_p \ln \frac{T_4}{T_2} - R_g \ln \frac{p_4}{p_2}$$

$$= 1.004\ \text{kJ/(kg}\cdot\text{K)} \times \ln \frac{1\,985\ \text{K}}{968\ \text{K}} - 0.287\ \text{kJ/(kg}\cdot\text{K)} \times \ln \frac{10.3\ \text{MPa}}{7.18\ \text{MPa}}$$

$$= 0.617\ 4\ \text{kJ/(kg}\cdot\text{K)}$$

平均吸热温度

$$T_{1\text{m}} = \frac{q_1}{\Delta s_{2-4}} = \frac{900\ \text{kJ/kg}}{0.617\ 4\ \text{kJ/(kg}\cdot\text{K)}} = 1\,457.6\ \text{K}$$

循环吸热量中的可用能：

$$e_{\text{x},Q} = \left(1 - \frac{T_0}{T_{1\text{m}}}\right) q_1 = \left(1 - \frac{293.15\ \text{K}}{1\,457.6\ \text{K}}\right) \times 900\ \text{kJ/kg} = 719.1\ \text{kJ/kg}$$

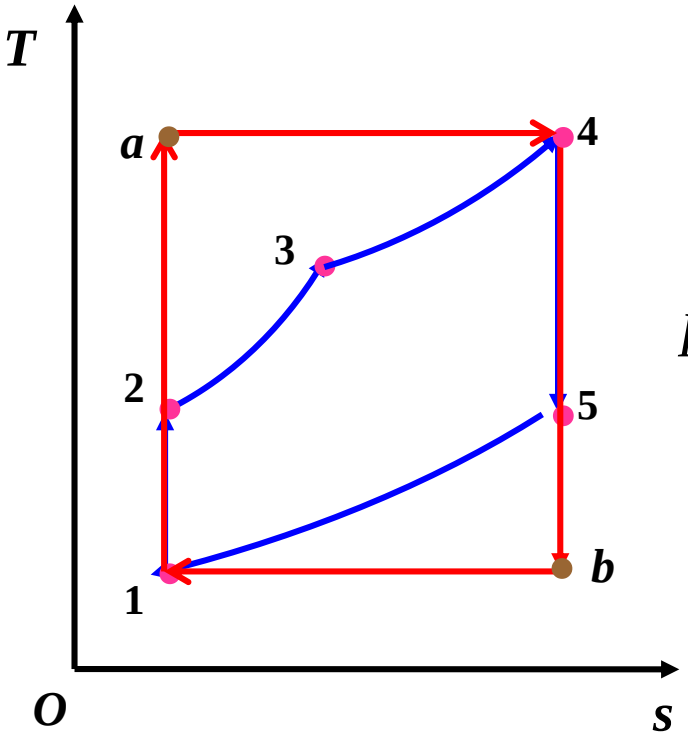
循环㶲效率

$$\eta_{e_{\mathrm{x}}} = \frac{w_{\mathrm{net}}}{e_{\mathrm{x},Q}} = \frac{580\ \mathrm{kJ/kg}}{719.1\ \mathrm{kJ/kg}} = 0.806$$

例题2：

若使活塞式内燃机按卡诺循环运行，并设其温度界限与下图所示混合循环相同，试从工程实践角度比较两个循环，已知：p_1 = 0.17 MPa，T_1 = 333 K，p_4 = 10.3 MPa，T_4 =1 985 K，v_2 = 0.038 7 m³/kg

解

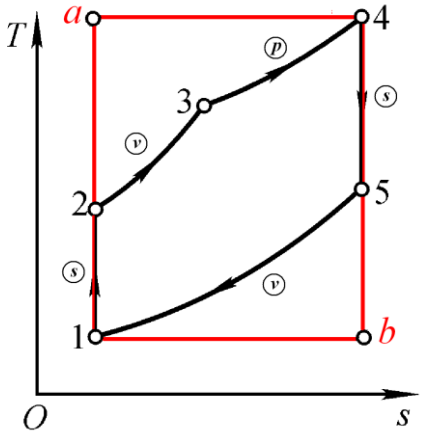


$$v_1 = \frac{R_g T_1}{p_1}$$

$$= \frac{287\ \text{J/(kg}\cdot\text{K)} \times 333\ \text{K}}{0.17 \times 10^6\ \text{Pa}} = 0.562\ \text{m}^3/\text{kg}$$

$$p_a = p_1 \left(\frac{T_a}{T_1} \right)^{\frac{\kappa}{\kappa - 1}}$$

$$= 0.17\ \text{MPa} \times \left(\frac{1\,985\ \text{K}}{333\ \text{K}} \right)^{\frac{1.4}{1.4-1}} = 87.9\ \text{MPa}$$



$$T_a = T_4 = 1\,985\ \text{K}$$

$$v_a = \frac{R_g T_a}{p_a}$$

$$= \frac{287\ \text{J/(kg}\cdot\text{K)} \times 1\,985\ \text{K}}{87.9\times10^6\ \text{Pa}} = 0.006\,48\ \text{m}^3/\text{kg}$$

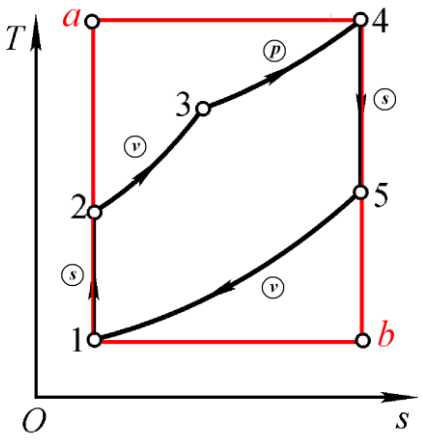
$$T_b = T_1 = 333\ \text{K}$$

$$p_b = p_4\left(\frac{T_b}{T_4}\right)^{\frac{\kappa}{\kappa-1}} = 10.3\ \text{MPa} \times \left(\frac{333\ \text{K}}{1\,985\ \text{K}}\right)^{\frac{1.4}{1.4-1}} = 0.019\,9\ \text{MPa}$$

$$v_b = \frac{R_g T_b}{p_b} = \frac{287\ \text{J/(kg}\cdot\text{K)} \times 333\ \text{K}}{0.019\,9\times10^6\ \text{Pa}} = 4.80\ \text{m}^3/\text{kg}$$

$$\eta_c = 1 - \frac{T_b}{T_a} = 1 - \frac{333}{1985} = 83.2\% > \eta_t = 64.0\%$$

从工程实用角度出发：



1）p_a= 87.9 MPa，压力太高；

2）p_b= 0.019 9 MPa，压力太低，

真空状态难以维持；

3）循环气体体积比

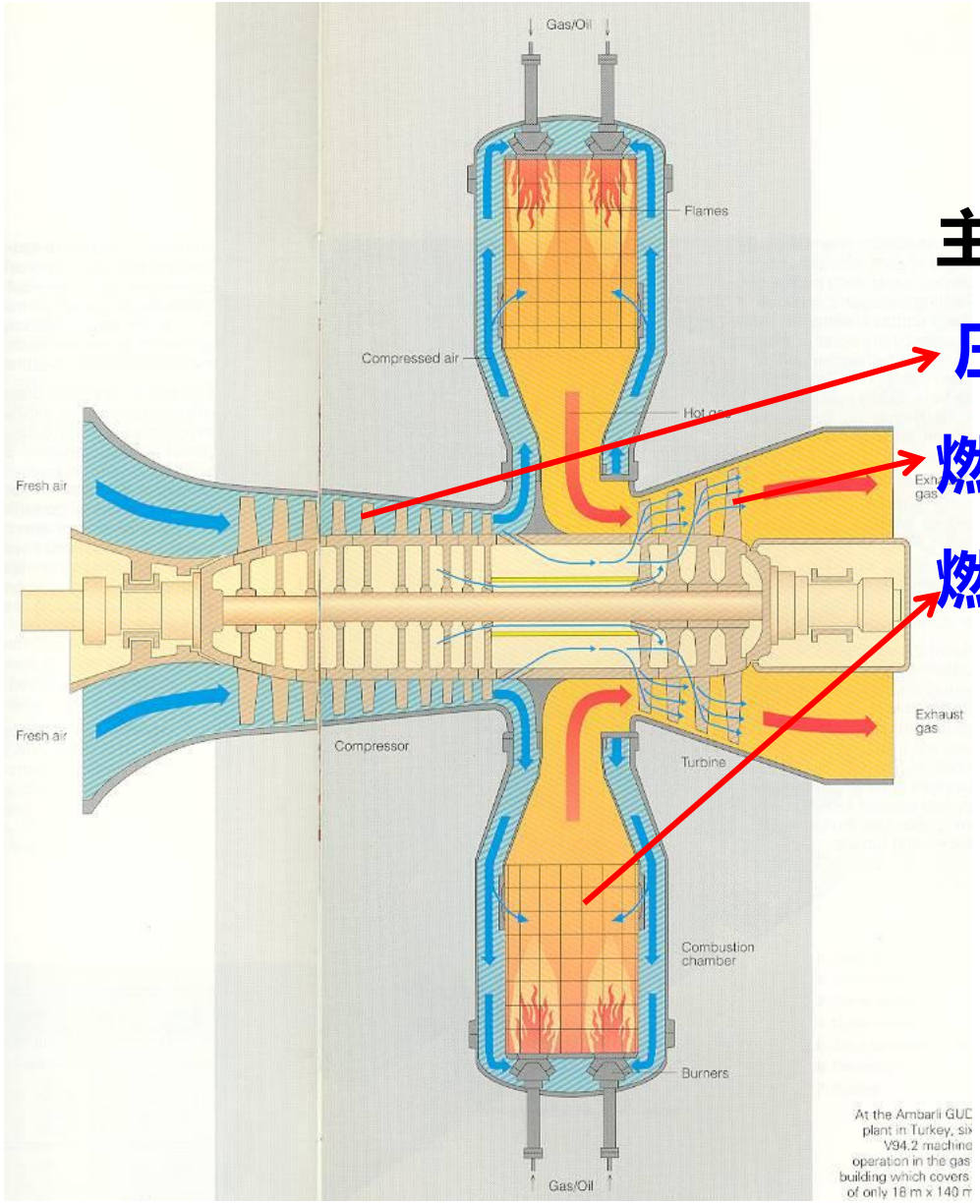
$$v_b = 4.8\mathrm{m}^3/\mathrm{kg} \quad v_a = 0.00648\mathrm{m}^3/\mathrm{kg} \quad \frac{v_b}{v_a} = 740.7$$

$$v_1 = 0.562\mathrm{m}^3/\mathrm{kg} \quad v_2 = 0.0387\mathrm{m}^3/\mathrm{kg} \quad \frac{v_1}{v_2} = 14.5$$

体积变化太大

4）活塞从$a \to b$移动距离太长，摩擦损耗太大，虽然每循环理论w_{net}增大，但实际收效甚少。

6–5 燃气轮机装置循环

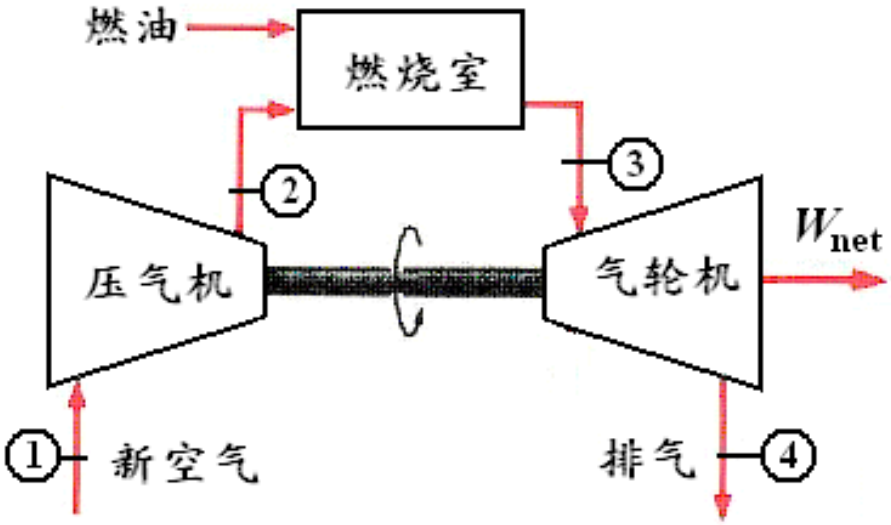


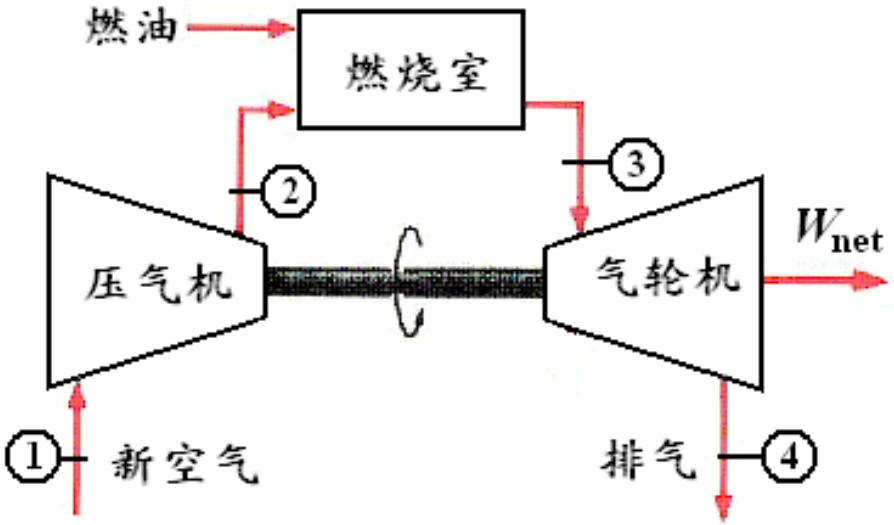
主要构成

压气机(compressor)

燃气轮机(gas turbine)

燃烧室(combustion chamber)





特点:

1. 开式循环(open cycle)，工质流动；
2. 运转平稳，连续输出功；
3. 启动快，达满负荷快；
4. 压气机消耗了燃气轮机产生功率的绝大部分，但功率重量比(specific weight of engine)仍较大。

用途

飞机、舰船动力载荷机组，电站峰荷机组(peak-load set) 等。

一. 运用空气标准假设简化

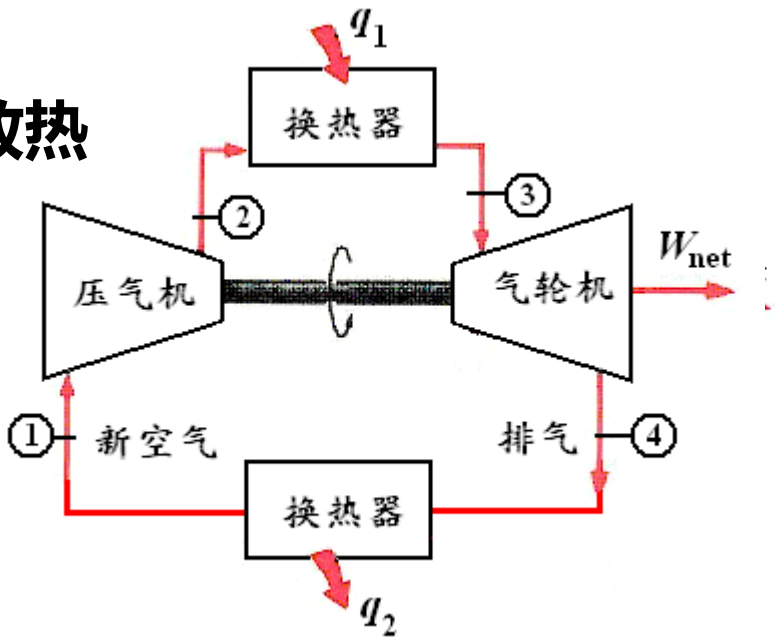
▲ 工质→空气，理想气体状态，定比热容

▲ 压缩、膨胀→等熵过程

▲ 燃烧→等压吸热；排气→定压放热

▲ 燃油质量→忽略

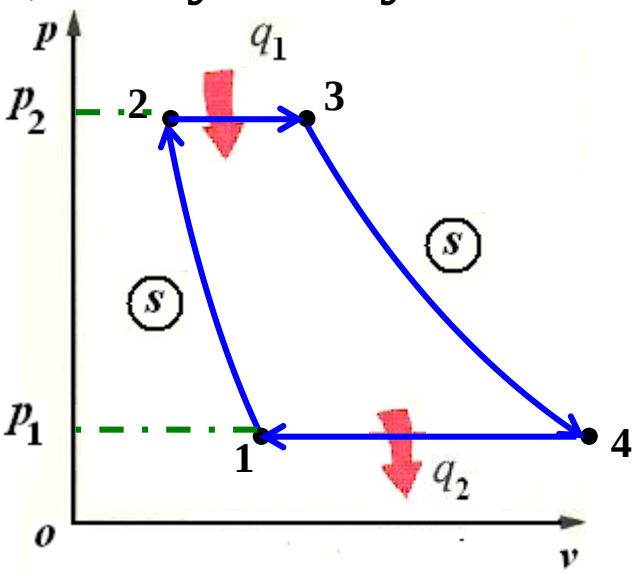
▲ 燃气成分改变→忽略



闭式定压加热循环

二、定压加热理想循环/布雷顿循环

（constant-pressure combustion cycle, Brayton cycle）



1-2 等熵压缩（压气机内）

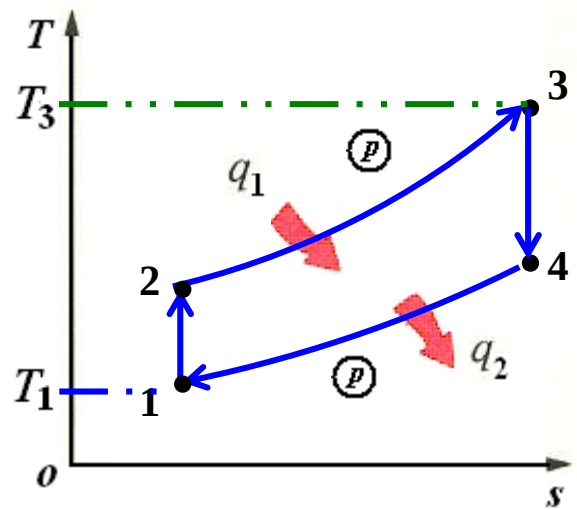
$$\pi = \frac{p_2}{p_1}$$

循环增压比（pressure ratio）

2-3 定压吸热（燃烧室内）

$$\tau = \frac{T_3}{T_1}$$

循环增温比（temperature ratio）

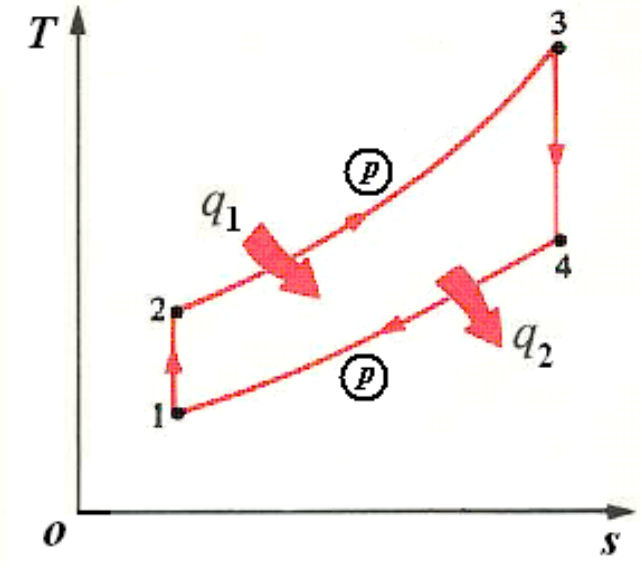


3-4 等熵膨胀（燃气轮机内）

4-1 定压放热（排气，假想换热器）

三、定压加热理想循环分析

1. 热效率η_t



$$q_1 = h_3 - h_2 = \bar{c}_p \Big|_{t_2}^{t_3} (T_3 - T_2) = c_p (T_3 - T_2)$$

$$q_2 = h_4 - h_1 = \bar{c}_p \Big|_{t_1}^{t_4} (T_4 - T_1) = c_p (T_4 - T_1)$$

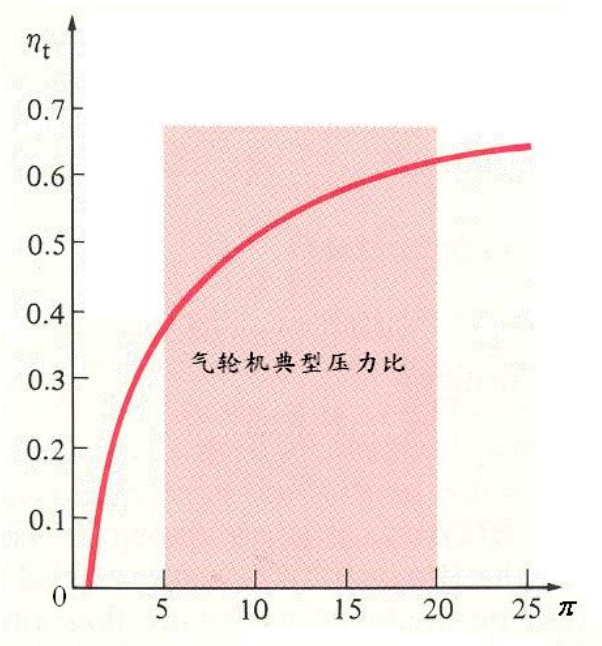
$$\eta_t = 1 - \frac{q_2}{q_1} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

$$\left.\begin{aligned} T_4 &= T_3 \left(\frac{p_4}{p_3}\right)^{\frac{\kappa-1}{\kappa}} \\ T_1 &= T_2 \left(\frac{p_1}{p_2}\right)^{\frac{\kappa-1}{\kappa}} \end{aligned}\right\} \quad \begin{aligned} p_4 &= p_1 \\ p_3 &= p_2 \end{aligned} \Rightarrow \frac{T_4}{T_3} = \frac{T_1}{T_2} \Rightarrow \frac{T_4 - T_1}{T_3 - T_2} = \frac{T_1}{T_2}$$

$$\therefore \eta_t = 1 - \frac{T_1}{T_2} = 1 - \frac{1}{\pi^{\frac{\kappa}{\kappa-1}}}$$

注意：式中T_1、T_2并非指高温热源，低温热源。

2. 分析 $\eta_t = 1 - \frac{T_1}{T_2} = 1 - \frac{1}{\pi^{\frac{\kappa}{\kappa-1}}}$



a) $\pi \uparrow$ $\eta_t \uparrow$

b) η_t**与循环最高温度** T_3**无关** ?

c) **π一定** $q_1 \uparrow$ η_t ?

$w_{net} \uparrow$ η_t**不变**

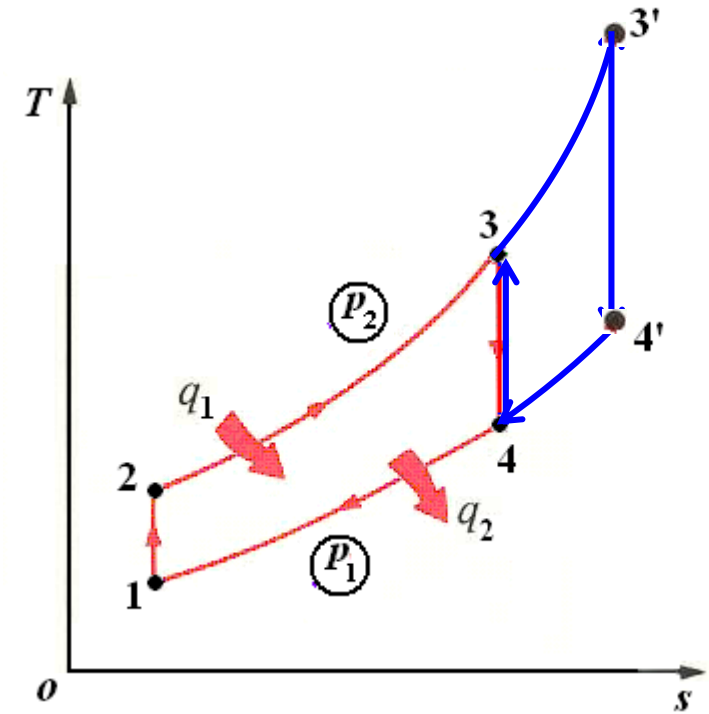
考察：循环123’4’1 =

循环12341 + 循环433’4’4

布雷顿循环 **布雷顿循环**

$\pi = p_2 / p_1$ $\pi = p_2 / p_1$

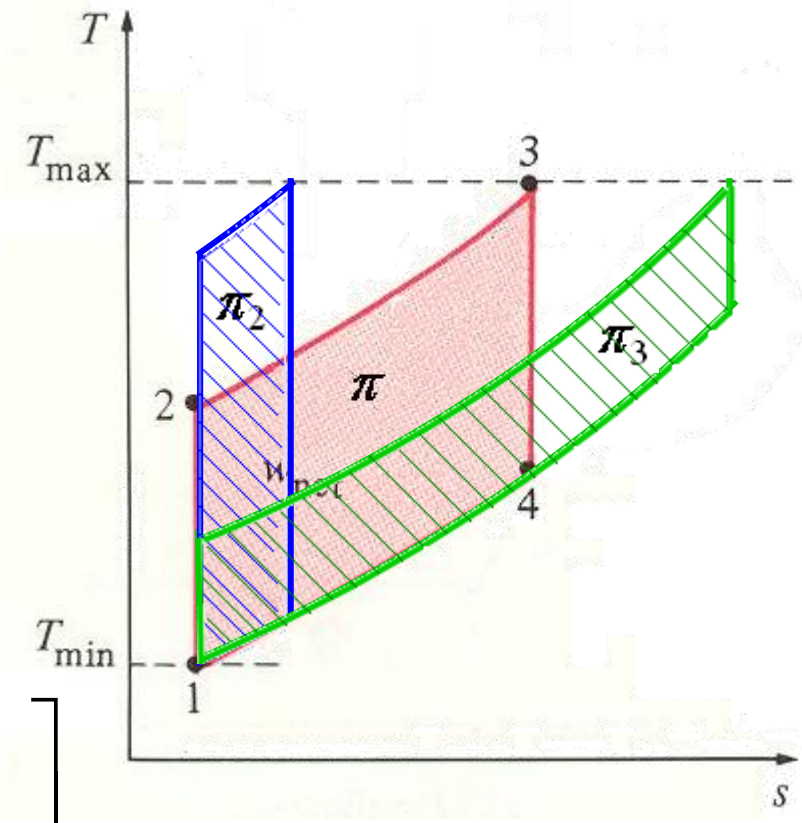
$$\eta_t = 1 - \frac{1}{\pi^{\frac{\kappa}{\kappa-1}}}$$



循环123’4’1热效率等于循环12341

d) τ一定，π取不同值

$$w_{\text{net}} \to w_{\max}$$



$$w_{\text{net}} = q_1 - q_2$$

$$= c_p\left[\left(T_3 - T_2\right) - \left(T_4 - T_1\right)\right]$$

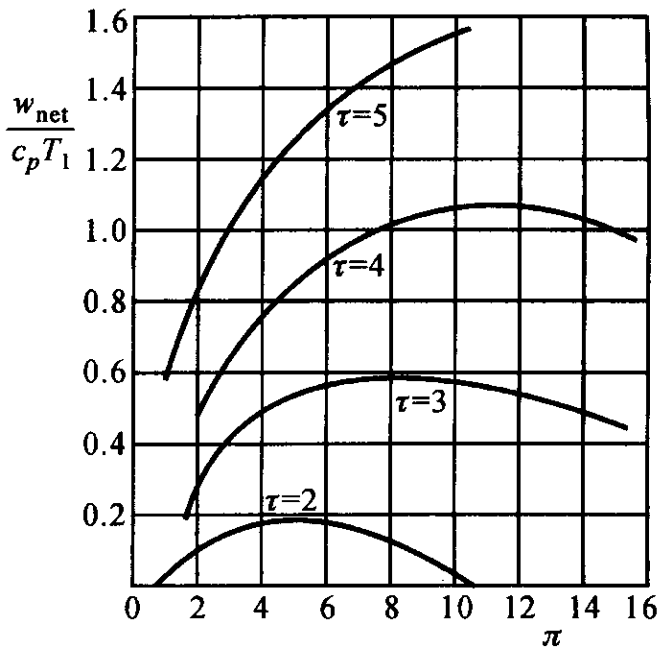
$$= c_p T_1\left[\frac{\tau\left(\pi^{\frac{\kappa-1}{\kappa}} - 1\right)}{\pi^{\frac{\kappa-1}{\kappa}}} - \left(\pi^{\frac{\kappa-1}{\kappa}} - 1\right)\right]$$

$$\frac{\delta w_{\text{net}}}{\mathrm{d}\pi} = 0 \longrightarrow \pi = \tau^{\frac{\kappa}{2(\kappa-1)}} \longrightarrow w_{\text{net}} \to w_{\text{net,max}}$$

e） w_{net} 与τ及π的关系

$$w_{net} = c_p T_1 \left[\frac{\tau\left(\pi^{\frac{\kappa-1}{\kappa}} - 1\right)}{\pi^{\frac{\kappa-1}{\kappa}}} - \left(\pi^{\frac{\kappa-1}{\kappa}} - 1\right) \right]$$

$$= c_p T_1 \left(\pi^{\frac{\kappa-1}{\kappa}} - 1\right)\left(\frac{\tau}{\pi^{\frac{\kappa-1}{\kappa}}} - 1\right)$$



1）对于每一τ，均有π，其$w \to w_{net,max}$

2）τ上升，即T_3上升，使取得$w_{net,max}$ 的π 上升，η_t上升，所以提高T_3 能带动$w_{net,max}$ 及η_t同时升高。

第6章小结

1、活塞式内燃机各种理想加热循环及各特性参数

2、燃气涡轮机装置循环及特性参数

3、循环的p-v图和T-s图

4、热效率计算

第6章作业

9-4，9-5

注意：只收手写作业！不收打印作业或电子版作业！